

When the cellular traffic changes, how much does an RL scheduler lose?

Quantifying transfer gaps in 5G RAN slicing

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1 The problem

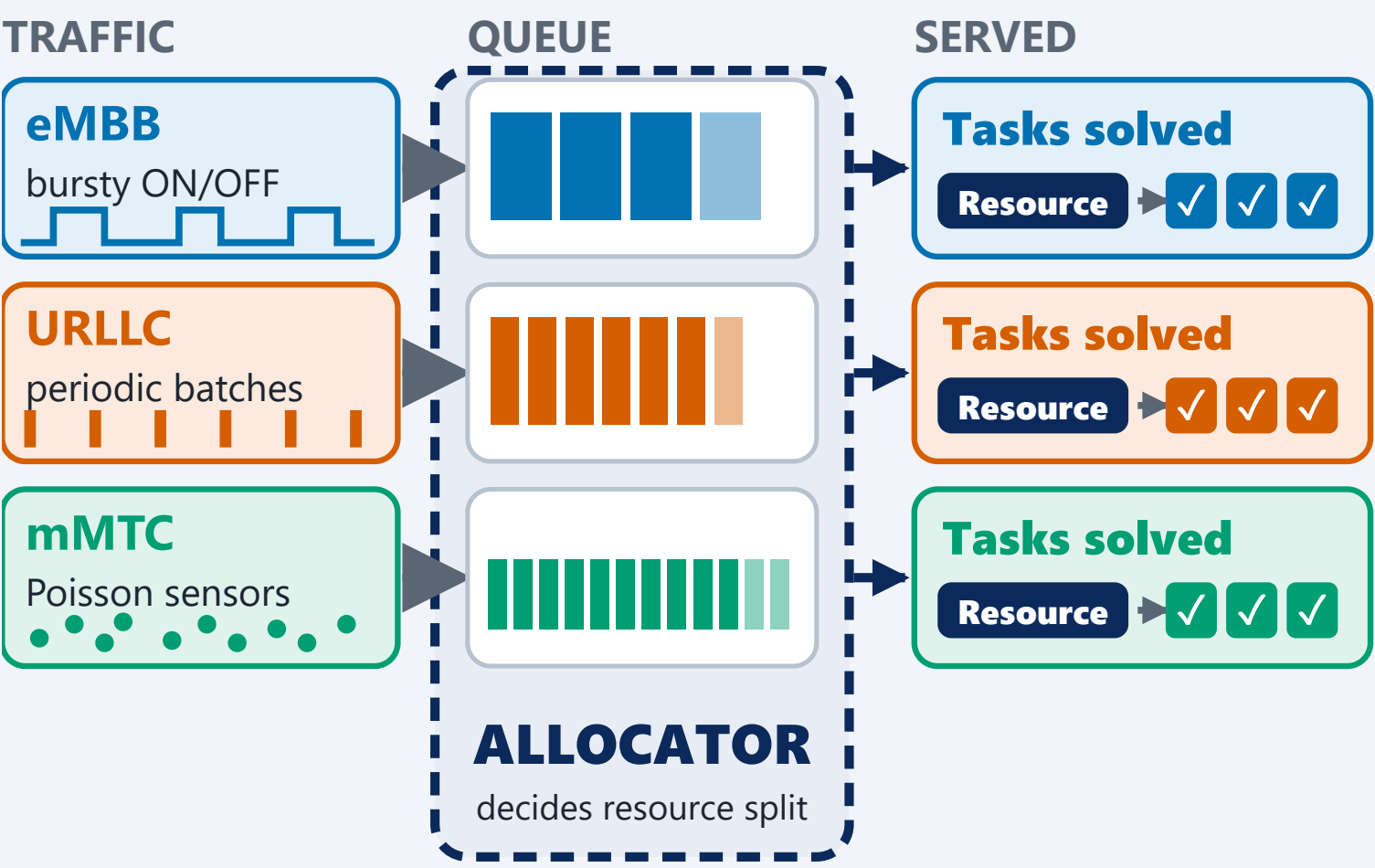
Every millisecond, a 5G base station must share its limited radio capacity between **three channels**, each carrying a different type of task:

- eMBB** video and downloads
- URLLC** robots and cars
- mMTC** sensor readings

Reinforcement learning (RL) is widely proposed to make this decision automatically.

► What if the traffic mix changes after training?

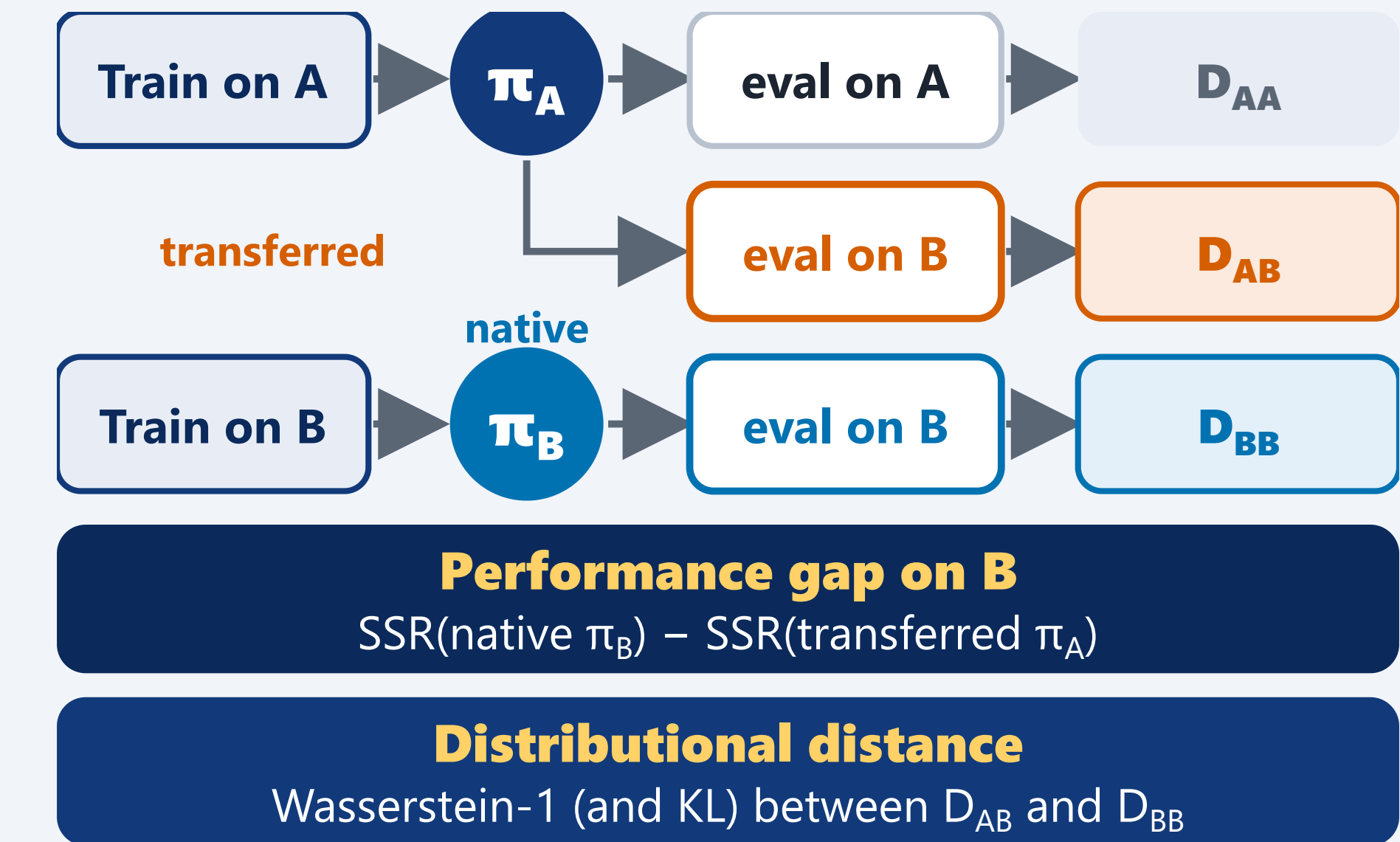
2 Simulator & four environments



Environment	Load share		
Balanced	33	33	33
eMBB-int	50	25	25
URLLC-int	25	50	25
mMTC-int	25	25	50

Same total load (96 %) everywhere: only the mix differs.

3 How we measure the loss



4 Reward & agent

$$R = \sum SSR - 0.1 \cdot \text{backlog}$$

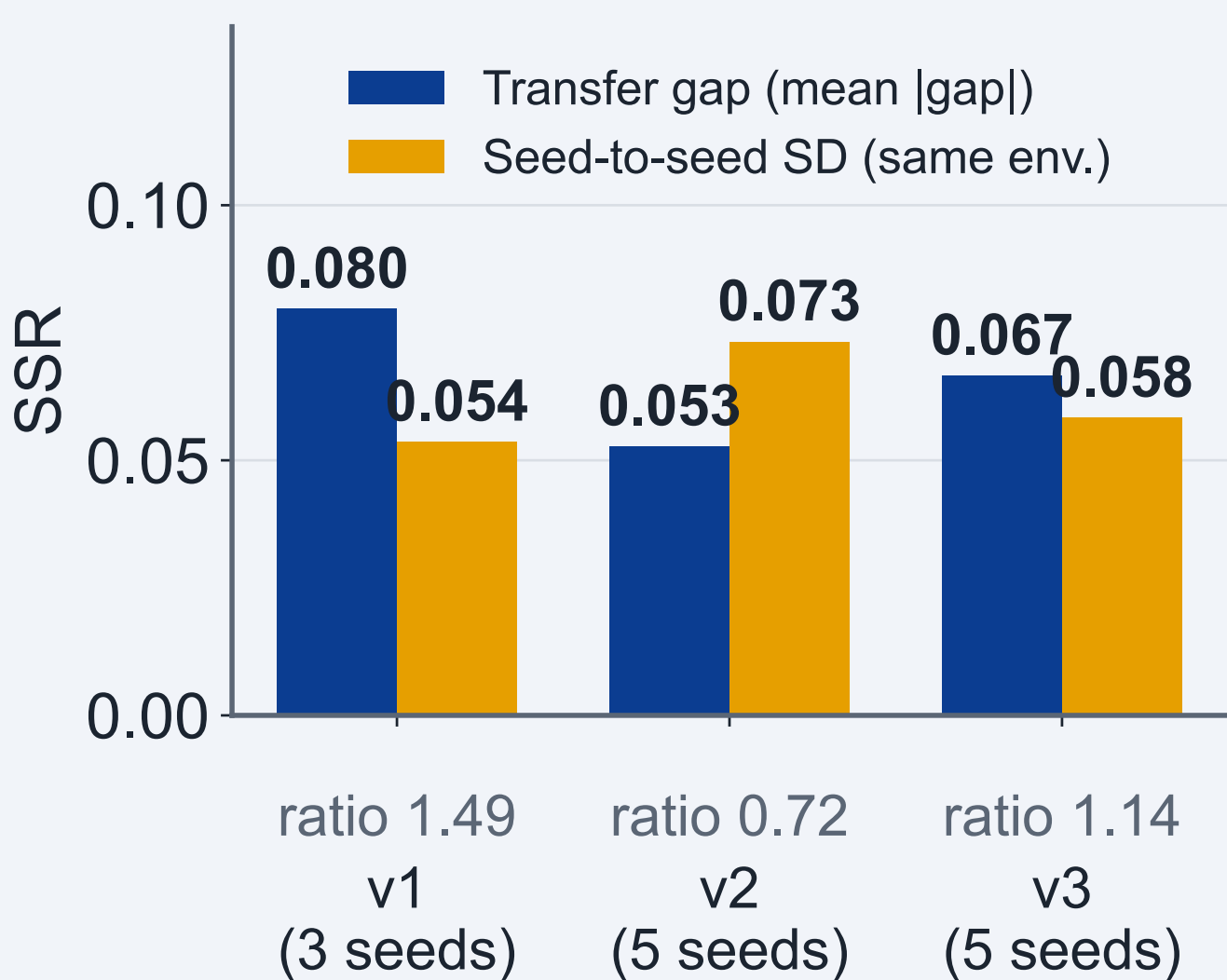
SSR = how well each channel's quality target is met (0 to 1). Rewarding it directly matches our evaluation metric.

- Agent** Deep Q-Network
- Seeds** 5 per environment
- Training** 600 episodes

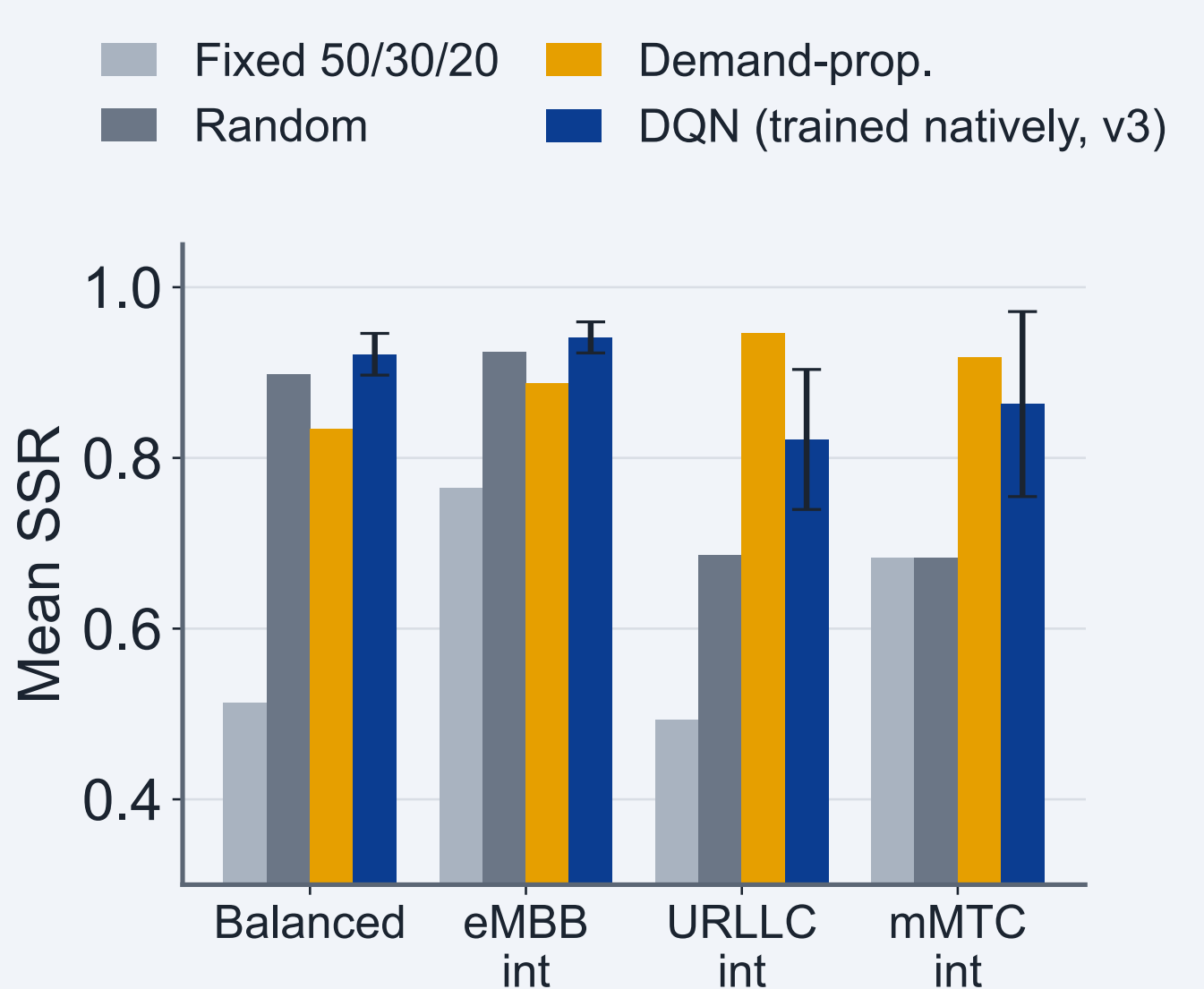
5 Results

	Evaluated on B			
	Balanced	eMBB int	URLLC int	mMTC int
Trained on A				
Balanced	native ref.	+0.016	-0.100*	-0.066
eMBB-int	+0.057	native ref.	-0.021	+0.107
URLLC-int	+0.061	+0.092	native ref.	+0.137*
mMTC-int	-0.014	-0.023*	-0.104*	native ref.

Gap by train (row) and test (column) environment



Transfer gap vs retraining with another seed



Mean SSR per environment

► Up to 0.14 SSR lost, yet many gaps are negative

► The gap is about as large as training noise

► RL beats a fixed split, not always simple heuristics

6 Conclusions

- Wrong traffic mix costs **5–8 SSR points**, similar to seed noise.
- Balanced training** transfers best.
- Distribution distance **does not predict** the gap.

Limits: single cell, DQN only. Next: PPO, mobility, more seeds.

Contributions

Mark Zhao

Built the simulator and RL agent; chose the parameters and assumptions

Stanley Findsen

Led the literature review and linked our work to current industry practice